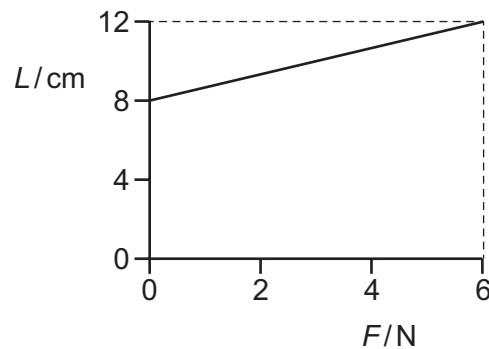


- 20** A spring is fixed at one end. The length L of the spring is increased by applying a tensile force F to the other end.

The graph shows the variation of L with F .



What is the elastic potential energy of the spring when F is 6.0 N?

- A** 0.12 J **B** 0.24 J **C** 0.36 J **D** 0.60 J
- 21** A wire has an unstretched length of 2.00 m.
- A stress of 1.6×10^5 Pa is applied to the wire, and the new length of the wire is 2.10 m.
- The wire obeys Hooke's law.
- What is the Young modulus of the wire?
- A** 8.0×10^3 Pa **B** 7.8×10^4 Pa **C** 1.5×10^5 Pa **D** 3.2×10^6 Pa
- 22** What are the SI base units of stress?

- A** kg m s^{-2} **B** $\text{kg m}^{-1} \text{s}^{-2}$ **C** $\text{kg m}^{-2} \text{s}^{-2}$ **D** $\text{kg m}^{-3} \text{s}^{-2}$